Pellet Printing for Soft Devices

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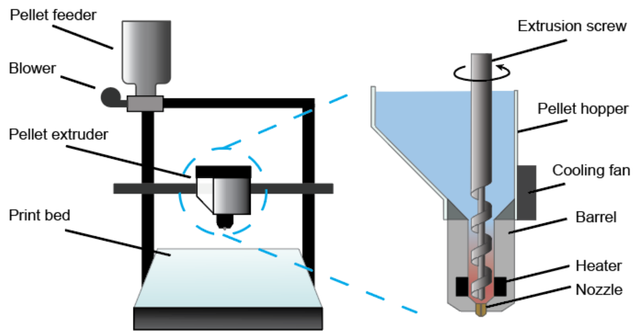
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Background and Motivation

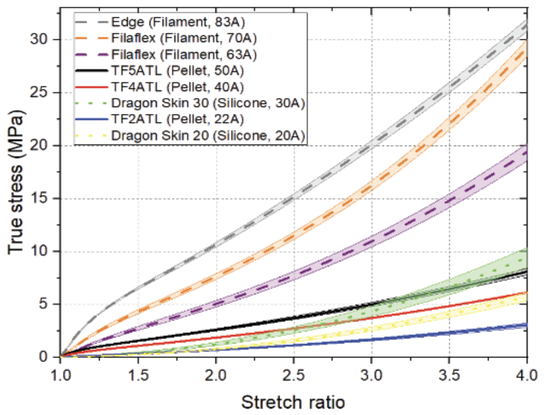
Soft robotics enables the creation of adaptable, safe devices such as medical tools, assistive wearables, and bioinspired robots [1]. Advances in soft materials and additive manufacturing have expanded the capabilities of these systems [1]. Most soft robots have been fabricated either via soft lithography or Fused Deposition Modeling (FDM). Soft lithography uses additively cured elastomers with low Shore hardnesses (00-10 to 00-60). FDM uses thermoplastics with significantly higher Shore hardnesses (60A to 95A). We explored Fused Granulate Fabrication (FGF) as a low-cost alternative that compares in simplicity to FDM and in Shore hardness of materials to soft lithography, a direction distinct from prior filament- and ink-based methods [2, 3].

Pellet Printer Development

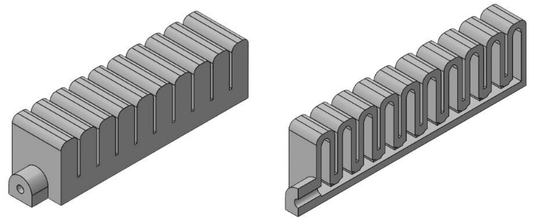
Our pellet printer was adapted from a desktop extrusion system to directly process thermoplastic pellets using a heated extrusion screw. Modifications, including a pressurized feeding system, optimized nozzle temperatures, and tuned screw speeds, enable reliable printing of ultra-soft materials.



I evaluated the performance of the printer through mechanical tests and demonstrations, fabricating soft robotic models to assess print reliability, material handling, and actuator durability.



PneuNet Design



A PneuNet is a pneumatic network, a soft robotic actuator made of internal chambers that expand and contract when air pressure is applied. For this study, our PneuNet had rounded channels and edges.

PneuNet Casting

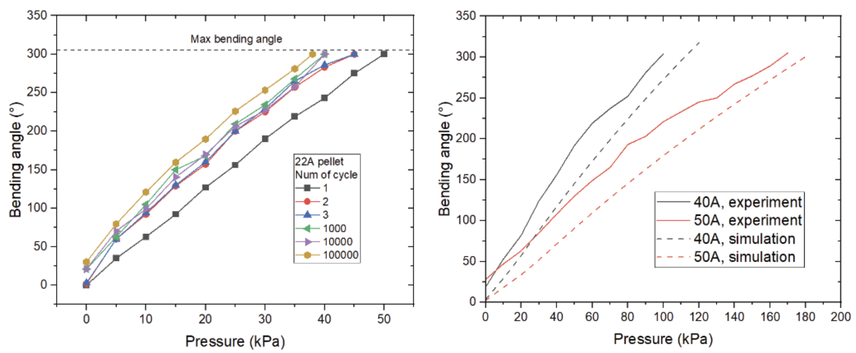


To create a benchmark test for comparative analysis between additively manufactured and soft lithography based pneumatic soft devices, we designed and fabricated custom molds of a PneuNet. Before casting the additive curing elastomer into the mold (Dragonskin 20A), the silicone was degassed using a vacuum chamber, eliminating possibly trapped air bubbles.

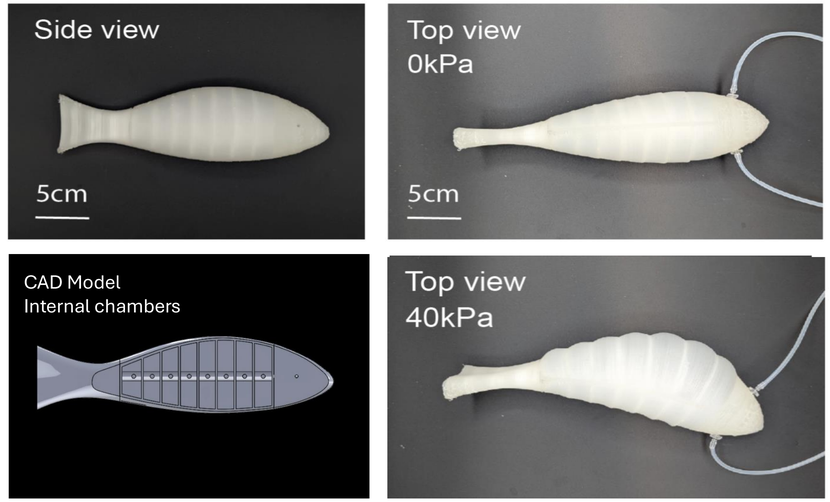
Cyclic Testing



Cyclic testing involved repeatedly inflating and deflating a PneuNet actuator from 0 to 50 kPa while measuring bending angles.



Soft Robotics Demonstration



Results

1. **Fabricating a soft device for benchmarking:** We successfully fabricated a PneuNet actuator from silicone using soft lithography and compared it with FGF-printed versions of the same actuator.
2. **Cyclic testing:** Our FGF-printed PneuNets maintained structural integrity over 100,000 cycles. Despite minor deformation, they were durable with no failure during continuous use.
3. **Demonstration:** We FGF printed a soft robotic fish with integrated pneumatic actuators, validating the ability of our pellet printer to fabricate complex, functional soft robots without post-assembly.

Acknowledgments

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Conclusion

Pellet-based 3D printing is a cost-effective option for the fabrication of soft devices. It bridges the gap between soft lithography and digital fabrication with thermoplastics of high Shore hardness. This work lays the foundation for broader, more accessible fabrication of silicone-soft robotic systems. Other 3D printing techniques are prohibitively expensive, require post-processing, or use viscoelastic materials.

References

1. T. J. Wallin, J. Pikul, R. F. Shepherd, 3D printing of soft robotic systems. Nature Reviews Materials 3 (6), 84–100 (2018), doi:10.1038/s41578-018-0002-2, https://www.nature.com/ articles/s41578-018-0002
2. T. J. K. Buchner, et al., Vision-controlled jetting for composite systems and robots. Nature 623 (7987), 522–530 (2023), doi:10.1038/s41586-023-06684-3, https://www.nature.com/ articles/s41586-023-06684
3. Y. Zhai, et al., Desktop fabrication of monolithic soft robotic devices with embedded fluidic control circuits. Science Robotics 8 (79), eadg3792 (2023), doi:10.1126/scirobotics.adg3792, https://www.science.org/doi/10.1126/scirobotics.adg3792.